Limits of Landauer and second law

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1. Introduction

This document is a supplement to the reference paper "Limits of Landauer principle and of the second law of thermodynamics" [1] [2]. It explains the simulation codes designed to demonstrate the article's conclusions.
We consider three types of memory inspired by the article by Jun et al.[3], which implement a particle evolving on a potential profile U(x,t) dependent on time t and abscissa x in a nanosystem.
- The device of Jun et al. is a bistable memory whose profile is tilted to force the particle to change state. We call it *tilting bistable memory*.
- A simplified bistable memory whose change of state is caused by a lateral shift of the profile. We call it *shifting bistable memory*.
- A memory with a single potential well whose change of state is caused by a translation of the well. We call it *shift memory*.

2. Potential energy landscape

The first scheme of the experiments is a double well potential landscape to realize a one-bit memory. A particle is represented by its position x and its potential energy. The phase space of the system is shown by a continuous curve f(x,U).
Our energy unit is $k_B\,T$, the entropy unit k_B, so the Clausius entropy inequation can be written $dS \geq dQ$.
Figure 1 shows a bistable memory with a symetric double well in state 1. The line separating the two states is conventionnaly at x = 0.

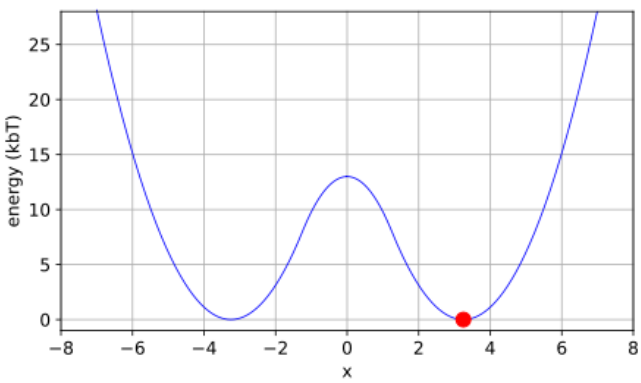

Figure 1 – Bistable memory in state 1

The following figure shows the energy transfers due to 3 causes : brownian mouvement, potential gradient $\frac{\partial}{dx}U(x,t)$ and action of actuator $\frac{\partial}{dt}U(x,t)$ changing the potential landscape.

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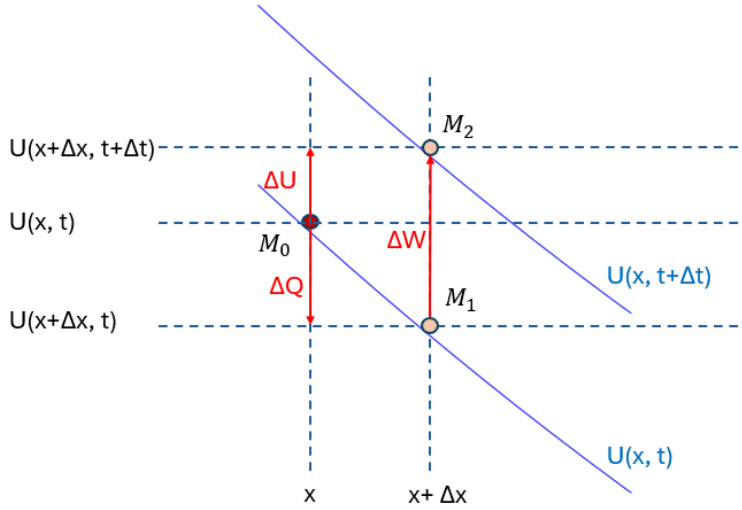

Figure 2 - Energy transfers during a period Δt

At the beginning of one period Δt, the particle is in M_0. During Δt, the particle moves by Δx from M_0 to M_1, due to potential gradient and to Brownian motion. An energy ΔQ is exchanged between the thermostat and the particle. At the end of the period, the energy landscape is suddenly modified from $U(x,t)$ to $U(x+\Delta x, t+\Delta t)$, exchanging energy ΔW between the actuator and the particle. The final balance is $\Delta U = \Delta Q + \Delta W$.

The finite difference equations are detailed in the reference paper. They allow to simulate the evolution of the memory from an initial state $x0$ and $U(x0, t0)$. Due to the high stochasticity of Brownian motion, the variables must be averaged over a large number of trials.

3. Tilting bistable memory

Jun et al. [3] made experiments to confirm Landauer's limit, which yielded highly accurate results. The energy landscape $U(x,t)$ is a quartic function of x:
$U(x,t) = Eb\,(x^4 - 2\,g(t)\,x^2 - 2\,f(t)x)$ with $Eb = 13$ (in $k_B\,T$ units)

phase	time interval	f(t)	g(t)
1 lowering barrier	$[0, tc/4]$	0	$1 - sin(\omega t)$
2 tilting	$[tc/4, tc/2]$	$4\,t/tc\ -\ 1$	0
2 raising barrier	$[tc/2, 3\,tc/4]$	1	$1 - sin(\omega(t - tc/4))$
4 untilting	$[3\,tc/4, tc]$	$4\,(1 - t/tc)$	1

with $\omega\ =\ 2\,\pi/tc$

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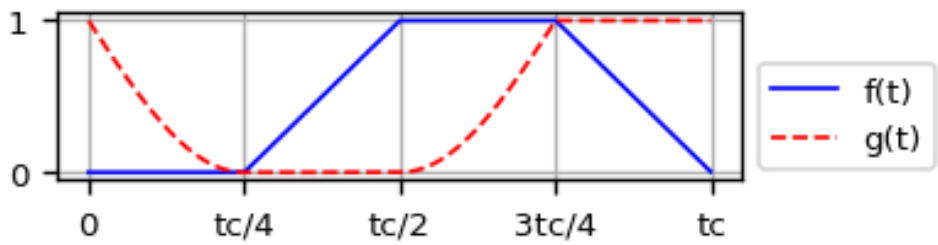

Figure 3 – Control functions f(t) and g(t)

The landscape at phase limits is given in figure 7 of the reference paper, and the evolution of the variables in figures 8 and 9.

4. Shifting bistable memory

In this case we limit the potential U between 0 and a maximum value of 13, with a process divided into 3 equal durations.To reach the same end value of work W = 0.71 we apply a cycle time of 1200 s instead of 940 s in the previous case.

Dynamic simulation of last phase.

To compute the dynamic evolution during the last phase, the algorithm based on the Langevin equation would require a very large amount of computing power.
So we apply a method based on the frequencies of state transitions as a function of their energy levels. It is not as accurate as it uses the Kramers formula which is approximate.
The potential profile of the right-hand well changes shape during the process, transitioning from a plateau $U = fim$ to a parabolic well, depending on the evolution parameter u, which varies linearly from 0 to 1 along with time t from 0 to tc = 400 s.

The transition frequency [1] F_{01} from state 0 to 1 is constant and is given by Kramers' formula: $F_{01} = f0\, exp(U_0 - U_b)$, where U_b is the potential of the barrier and U_0 is the average energy at state 0, which is constant, and:
$f0 = \frac{\sqrt{U_b''\, U_t''}}{2\,\pi\, D}$ with D = 2.5 (diffusion coefficient)
U_b'' and U_t'' are the second derivative in x of U(x,t) at well bottom and barrier top, which are :
$U_b'' = 21.6$ and $U_t'' = 108$
which gives $f0 = 2.0\ Hz$

Kramers formula does not apply at the beginning of the process when the right profile is a plateau or a shallow well. So we will compute F_{10} directly. Since we repeat the process a large number of times, we can use average values to obtain F_{10}.
We first calculate in terms of u the average energy of the particle when it is in state 1, using the formulas in Section 3 of the reference paper.

The segment $[0, xmax]$ is divided into N equal segments.
The average energy of state 1 is: $U_s(u) = \sum_{xi=0}^{xmax} P(x_i, u)\ U(xi, u)$
with $P(x_i, u)$ = exp$(-U(x_i, u))$ / $Z(u)$ and $Z(u) = \sum_{xi=0}^{xmax} \exp(-U(xi, u))$
We can apply the formula to compute the energy of state 0 (when the particle is in the left well) where u = 0: $U_0 = U_s(1)$
We find $U_0 = 0.5$ which is a characteristic of a parabolic well.

[1] More precisely, F_{01} is the inverse of the average dwell time in state 1. We refer to it as frequency for convenience.

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For a large number of experiments, the dwell times t_0 and t_1 of the particle in each state will be such that $t_0/t_1 = \exp(U_s(u) - U_0)$,
so we get: $F_{10} = F_{01} \exp(U_s(u) - U_0)$
The probability P follows the differential equation $dP/dt = F_{01} P - F_{10} (1 - P)$
or $dP/du = (F_{01} P(u) - F_{10} (1 - P(u))) t_c$
Knowing the initial value $P(0) = 0$, we can compute the evolution $P(u)$ by finite differences, and get the work W by $dW = P(u) \frac{dU_s(u)}{du} du$
The result for several durations of the experiment is given on figure 10 of reference paper.

5. Shift memory

Using the same equipment employed to create the bistable memory — a colloidal particle in a fluid — we can design a single well memory in which the information entropy is controlled and stable, without the need for a potential barrier.
The application of the algorithms of the previous section are straightforward.
The results show that the work necessary to reverse the state from 0 to 1 is generally lower than 0.05 kBT for a cycle time higher than 100 s.

Example of results for ns = 1000 simulations and tc values in seconds.
Due to stochasticty, the results vary for every computation.

tc	W
10	4.568
20	0.993
30	0.109
40	0.270
50	0.118
60	0.108
70	**0.051**
80	0.034
90	0.006
100	0.020

Influence of the number of experiments on W for tc = 70 s

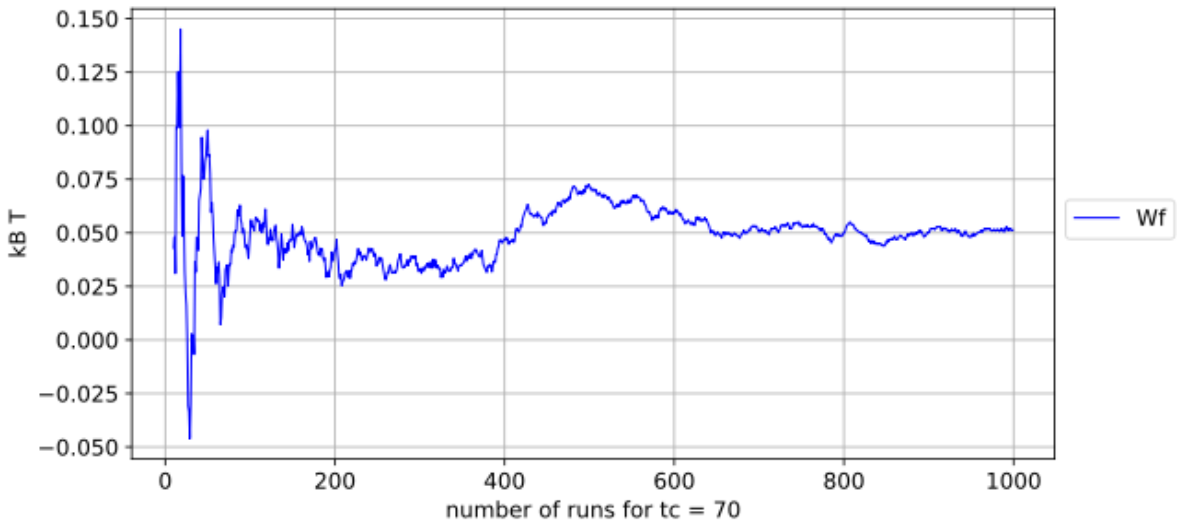

Figure 4 – Average work necessary for a NOT operation
as a function of number of runs.

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Bibliography

[1] J. Argouarc'h, « Limits of Landauer's principle and of the second law of thermodynamics », *Entropy: Thermodynamics – Energy – Environment – Economy*, vol. 7, nº Issue 1, 2026, doi: 10.21494/ISTE.OP.2026.1414.
[2] J. Argouarc'h, « Limites du principe de Landauer et du second principe de la thermodynamique », *Entropy: Thermodynamics – Energy – Environment – Economy*, vol. 7, nº 1, 2026, doi: 10.21494/ISTE.OP.2026.1413.
[3] Y. Jun, M. Gavrilov, et J. Bechhoefer, « High-precision test of Landauer's principle in a feedback trap », *Phys. Rev. Lett.*, vol. 113, nº 19, p. 190601, 2014.

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